# Fullsoul Insoles Sensing device, and its practical applications.

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**Abstract**

For most of human history, people simply walked barefoot. Shoes are a relatively recent invention—appearing roughly 40,000 years ago, and over time they evolved into the cushioned, structured models we know today. For a long time, that development was seen as straightforward progress. Yet in the late 2000s, with the rise of interest in minimalist and barefoot-style footwear, this assumption began to be questioned.

Despite this, relatively little is known about a basic issue: does the material of the sole actually make a difference? If the thickness is kept at a minimum, simulating the sensory and mechanical characteristics of barefoot walking, would it be different to using a traditional shoe? Although this consideration may appear trivial at first glance, its importance should not be underestimated. Subtle variations in the mechanical and material properties of the sole can influence the distribution of gravitational forces throughout the body, with far-reaching implications for both comfort and musculoskeletal health. Accordingly, this paper aims to explore the different mechanics of movement, to gain useful information on the difference of walking between these two different concepts.

## 1 Introduction

This project is a further development of a 2024 university project focused on wireless plantar pressure measurement using sensor insoles, initially conducted by Burian et al., 2024. The system captures real-time pressure data from 208 sensor channels embedded in shoe insoles, transmits it to a laptop, where the

data is then stored in HDF5 format (or CSV/SQLite), and visualizes it interactively as pressure heatmaps for applications such as gait analysis or medical diagnostics. The hardware is based on, Velostat-based pressure-sensitive sensors and a custom multiplexer PCB.

This project was not easily done. As a matter of fact, this project was plagued from the start by numerous issues, which naturally cut our time actually gathering data. The implementation of the process required expertise in the field of welding and soldering, because of a vital part. The battery, to be precise. The cables that connected it to the motherboard continued to snap, rendering the process extremely slow. We have also decided to switch off the BLE (Bluetooth Low Energy), preferring to return to a cable based connection. Despite this, this paper will present an experimental analysis of gait, based on experimentation, and deliver replicable data. To start an analysis, however, one must look at the past of the topic; therefore it is appropriate to start by delivering a review of the literature that brought the group to this point.

## 2 Literature review

As stated in the abstract, the challenging of the paradigm of exercising barefoot is a recent concept. This poses a significant challenge trying to systematically review the literature. Indeed, as pointed out by table 1 of Hollander's et.al. *Long-Term Effects of Habitual Barefoot Running and Walking: A Systematic Review*, we can see that out of the 15 highlighted studies, 8 have been conducted during/after 2010, and out of those 8, 5 have been authored during/after 2015[1]. As it being a recent field, it also stands to be a field in which long-term evidence gathering is still a test. Therefore, our research focuses on the evidence gathering stage.

## 3 Initial State and Challenges

As previously noted, this project was not without challenges. The initial hardware and codebase were not delivered in a satisfactory state, and the preceding team's documentation of their progress—including details of the Python environment and associated dependencies—was incomplete. This is reported here not as criticism, but to clarify the constraints inherited at the outset of the work; these factors directly contributed to the limited documentation and reduced transparency encountered during the project.

Specifically, the wireless communication system implemented by the previous group relied on an ESP32-C3 microcontroller for Bluetooth Low Energy (BLE) communication. This system was intended to transmit measurement data wirelessly from the insole to a laptop via BLE, a feature that would have been highly advantageous, as it would have facilitated the use of the insole during experimental trials. However, the laptop used in this project was unable to reliably

[1] page 4 & 5[1]

detect the insole, and in those instances where a connection was established, the system failed to record valid data. This might have been more readily resolvable had the software environment and configuration been thoroughly documented, but this was not the case.

Troubleshooting this issue required approximately one month. During this period, the software repeatedly generated connection errors that were difficult to interpret. Subsequently, it was determined that certain software components deployed on the BLE device were not fully compatible with Windows 11—information that had not been communicated in the available documentation.

In addition, the cable connecting the ESP32-C3 to the motherboard required repeated inspection and re-soldering due to intermittent connectivity problems.

Another recurring challenge concerned the Velostat sensor layer, i.e., the pressure-sensitive material forming the measurement matrix. This layer exhibited a minor tear in the conductive path between the motherboard and the sensing layer. A temporary repair was performed using adhesive tape; however, a robust permanent repair is still needed. Implementing such a repair would require specialized technical expertise that exceeded the capabilities and resources of the present team.

These persistent difficulties forced the team to switch back to a wire -based connection, instead of a wireless transmission.

# 4 Collaborators

The data were obtained from experiments conducted by the authors, which are detailed as follows:

- male (1) — 26 years old. — 79 kgs.— 182 cm.
- male (2) — 26 years old. — 71 kg —182 cm.
- male (3) — 29 years old — 94 kg. — 184 cm.
- female (1) — 27 years old — 49 kg. — 153 cm.

This was done to prevent ethical blocks. The data was also recorded in one go on february 19 2026. This is not mentioned in passing, as the weather will have an impact as to the outcomes of the experiments. This will however be explained later.

# 5 Experiment description

We have decided to conduct three experiments in total, the first one being an experiment conducted on a flat surface; the second was conducted on stairs, and the third was conducted on a slope. Also, as our hypothetical is to figure out if walking on a flat sole rather than a gumshoe [2]sole would impact our weight

[2]In this text, the words gumshoe and rubber sole will be used interchangeably, as the meaning is aproximately the same.

distribution and plantar health, the experiments were then repeated twice, once on a leather shoe called Fullsoul, which does not possess the gumshoe sole, and then with our own shoes.

### 5.1 flat-surface experiment

for the flat surface experiment, the following parameters were chosen: we would first start on the right foot in the Fullsoul, and we would walk on a straight line in a hall for twenty steps. This procedure was repeated by every collaborator. After the recording on the right was done, the same thing was done on the left.

### 5.2 stair-experiment

the stair experiment essentially consisted in walking up the stairs while being wired in, for twenty steps. Once the twenty steps were done (this usually consisted in walking up two stories, as a reference of the distance), the experimenter would then walk down the same flight of stairs. This was done, as previously mentioned, for both the left foot, the right foot; and for the Fullsoul and the regular shoe.

### 5.3 slope experiment

the experiment with the slope was done slightly differently. We found a straight sloped path of at least 5% and then the experimenter would walk in a normal gait for 12 steps down the slope. Once that data is recorded, the experimenter would turn and then walk up the slope for 12 steps. This was done for both right and left foot; and was made possible through the use of Fullsoul, and the the regular gumshoe.

Once the data was collected, our next steps were to rename and re-organize them into different main directories, for ease of collection.

## 6 Visualization

the raw HDF5 were then put through a visualizer which would create frames of the overall heatmap. These collection of frames would the be re-worked into a proper video. These videos, which were also aided by video recording of the experimenter would then be compiled in a side by side view to help aid with the visualization. An example of this is shown in picture one, which is a snapshot of the stairs-experiment of examiner Francesco. This snapshot is particularly useful because it allows us to pivot to talk about the findings of these experiments.

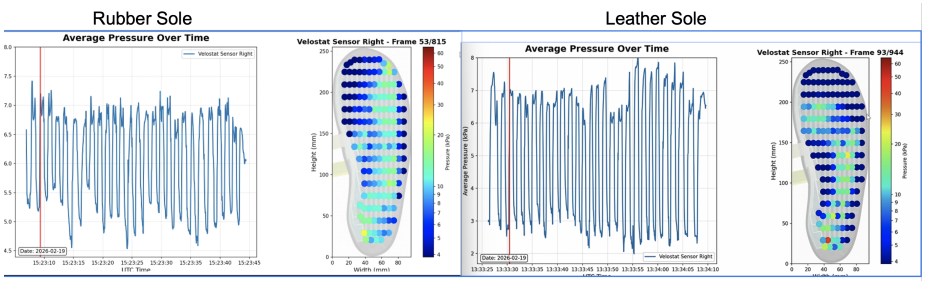


Figure 1: Rubber vs. Leather Shoe comparison

# 7 Findings

Remaining on the previously discussed experiment. While walking **up** the stairs, there appears to be a more evenly weighted distribution of pressure across almost the entire foot for the **rubber shoe.** This suggests that the foot, while being in a rubber sole, the foots loads pressure more uniformly rather than selectively. This might seem a good thing at first glance, however, it is actually unhealthy, as selective weight distribution allows the foot to distribute more weight in certain parts, which are naturally more efficent at taking the pressure load. The **leather sole** tells a different story, the weight distribution appears more curved and concentrated along what would be considered the healthier, more natural load-bearing areas of the foot. Another thing which was noted during the experimentation was that the pressure changes more frequently when using the leather sole. There seems to be more variation and oscillation in pressure, suggesting a dynamic adjusting of the weight while moving, which makes sense. In contrast, the regular **gum-shoe** shows not only less fluctuation of the pressure, but a higher pressure overall. The same pattern previously described show up while also completing the second part of said exercise (going down the stairs), and generally, we see the same in all experiments we have conducted.

# 8 Recommendations for future work

The following improvements and repairs are recommended for the next team to ensure reliable and long-term usable measurements.

- **Repair of the damaged Velostat layer**: The connection between the Velostat foil and the ESP32 board would have to be repaired. A proper, permanent repair is strongly recommended. This requires electronic skills (fine soldering of traces or replacement of the damaged section). Without this repair, measurement reliability over longer periods cannot be guaranteed.

- **Revival or replacement of wireless BLE functionality**: Obviously, the revival of the BLE might be a relevant area for further study. This

also includes clarifying whether the issue is limited to a hardware.

- **Improvement of visualization** Add center-of-pressure (CoP) path tracking. Optimize video export (MP4 instead of HTML-only). Implement automatic scaling and better interpolation for smoother heatmaps.

# References

[1] Karsten Hollander, Christoph Heidt, Babette C Van der Zwaard, Klaus-Michael Braumann, and Astrid Zech. Long-term effects of habitual barefoot running and walking: a systematic review. *Medicine & Science in Sports & Exercise*, 49(4):752–762, 2017.